

# my\_fitpca

Create the my\_fitpca function that we created in the lecture. It takes as input the [MxN] data matrix **D**, containing N feature measurements for each sample in the [Mx1] vector **class**. The function outputs the struct **mdl**, which contains the pca models created for each class.

## Function ?

Save

Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```

1 function mdl = my_fitpca(D,class)
2 % Find the number of unique classes from the class vector, numclasses
3 classes = unique(class);
4 numclasses = size(classes,1);
5 % For each class, use 'pca' to find the eigenvectors (1st output), eigenvalues (3rd output), and mean feature
6 %mdl = struct('class',[1]);
7 for i=1:numclasses
8     cl = classes(i);
9     [eigenvectors,~,eigenvalues,~,~,mu] = pca(D(class==cl,:));
10
11     % create a subfield 'class' within the output struct mdl, which is a vector of length numclasses
12     % mdl.class(i) has subfields 'eigvects' (the eigenvectors matrix output from 'pca', transposed), 'eigvals'
13     mdl.class(i) = struct('eigvects', eigenvectors', ...
14                          'eigvals', eigenvalues, ...
15                          'mu', mu);
16 end
17 end

```

## Code to call your function ?

Reset

```

1 D = [1 3 2 4;1 3 4 5]';
2 class = [0;0;1;1];
3 mdl = my_fitpca(D,class);
4 mdl.class(1) % eigvects: [0.7071 0.7071]; eigvals: 4; mu: [2 2];
5 mdl.class(2) % eigvects: [0.8944 0.4472]; eigvals: 2.5000; mu: [3 4.5000];
6
7
8 D = [1 3 2 4 5 ;1 3 4 5 6; 6 3 2 1 7]';
9 class = [0;0;1;1;2];
10 mdl = my_fitpca(D,class);
11 mdl.class(1) % eigvects: [-0.4851 -0.4851 0.7276]; eigvals: 8.5000; mu: [2 2 4.5000]
12 mdl.class(2) % eigvects: [0.8165 0.4082 -0.4082]; eigvals: 3.0000; mu: [3 4.5000 1.5000]
13 mdl.class(3) % eigvects: [0x3 double]; eigvals: [0x1 double]; mu: [5 6 7] (We get empty eigvects and eigvals)
14

```

Run Function

?

## Assessment: All Tests Passed

Submit

?

✔ All sub-fields correct for random valued input?